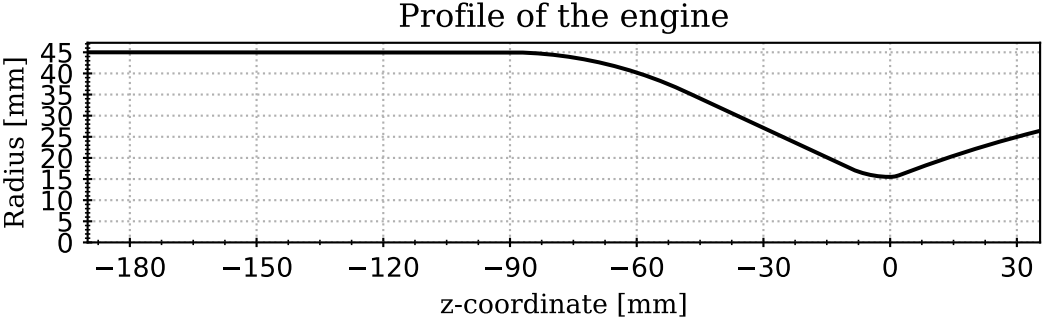
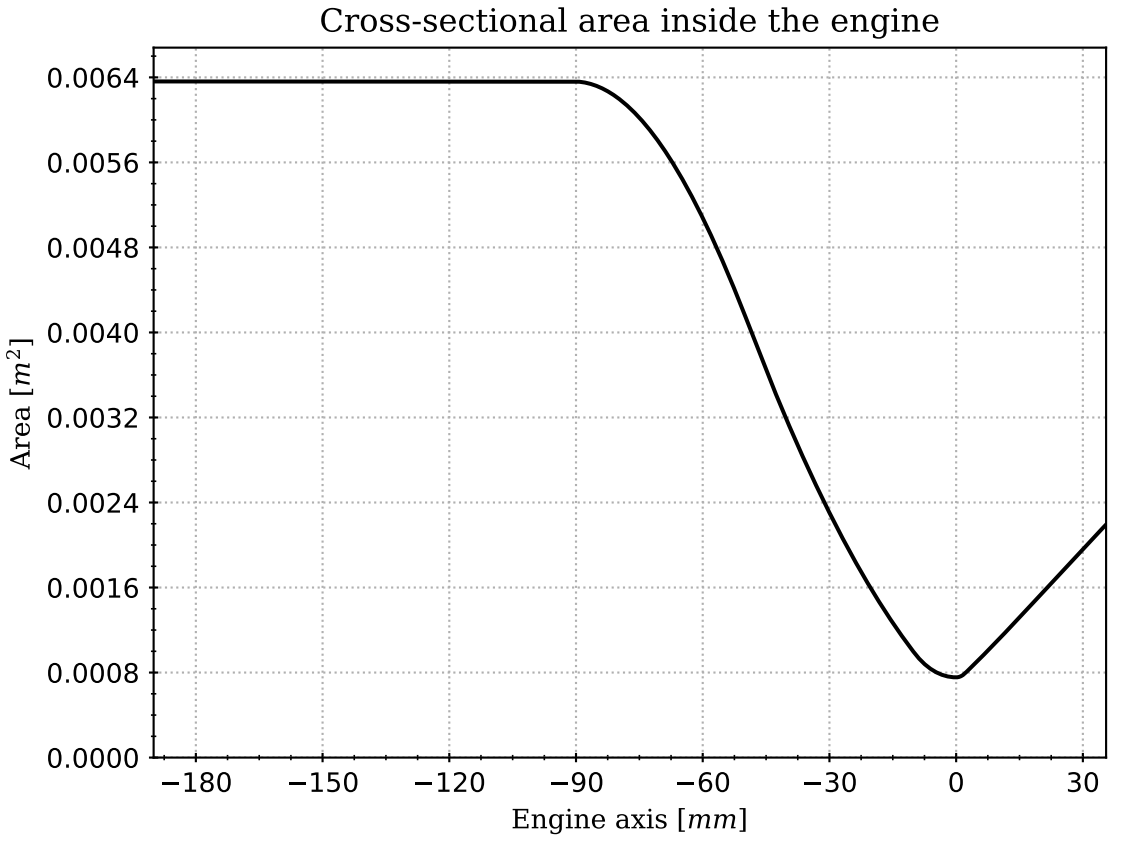
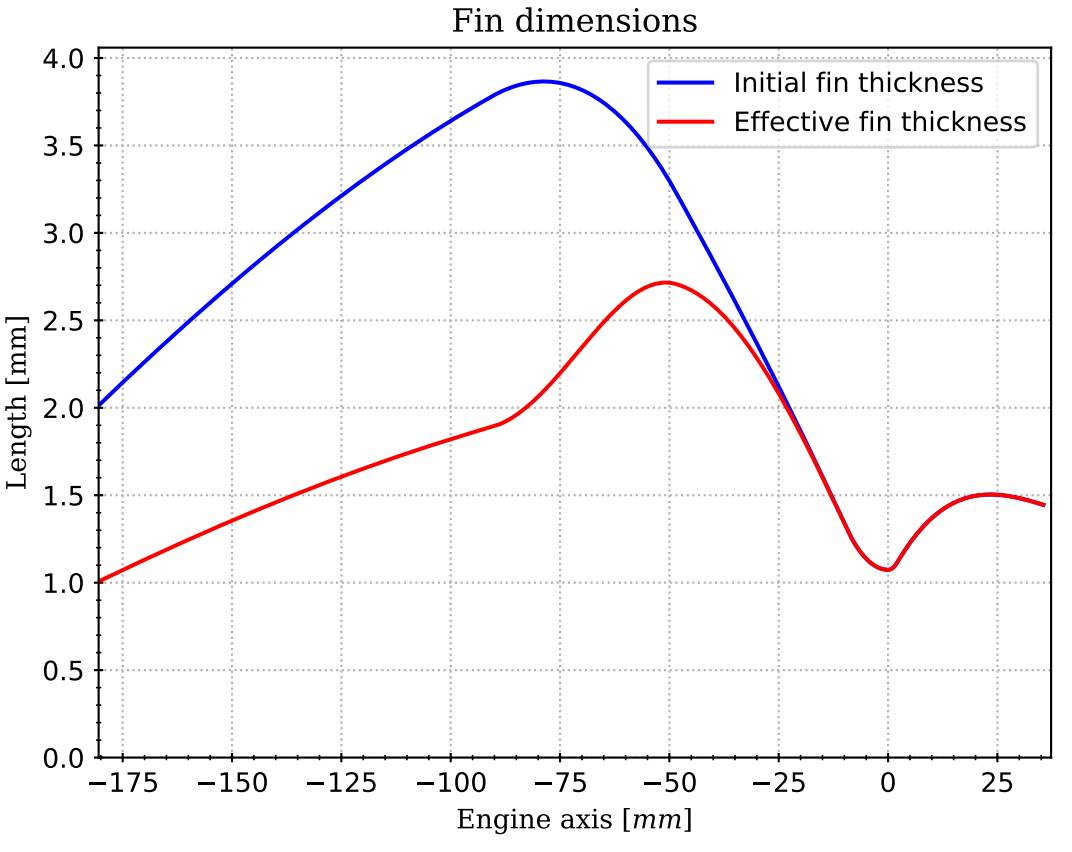
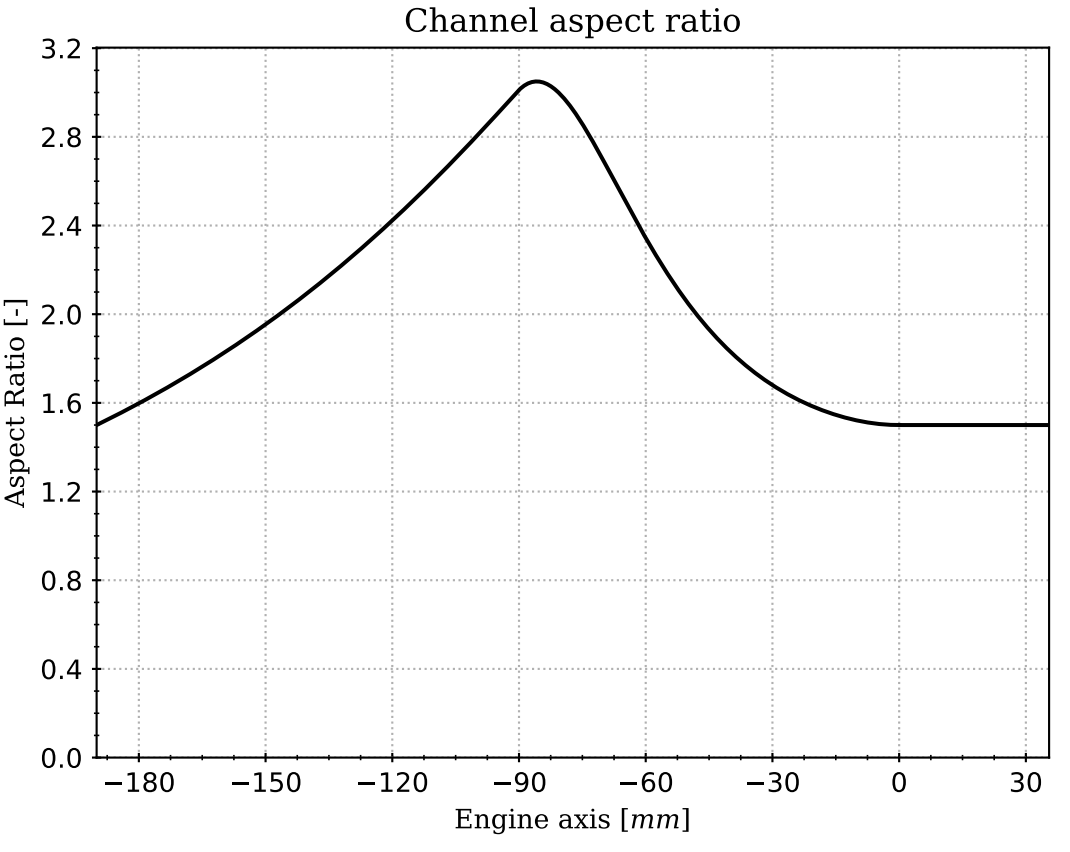
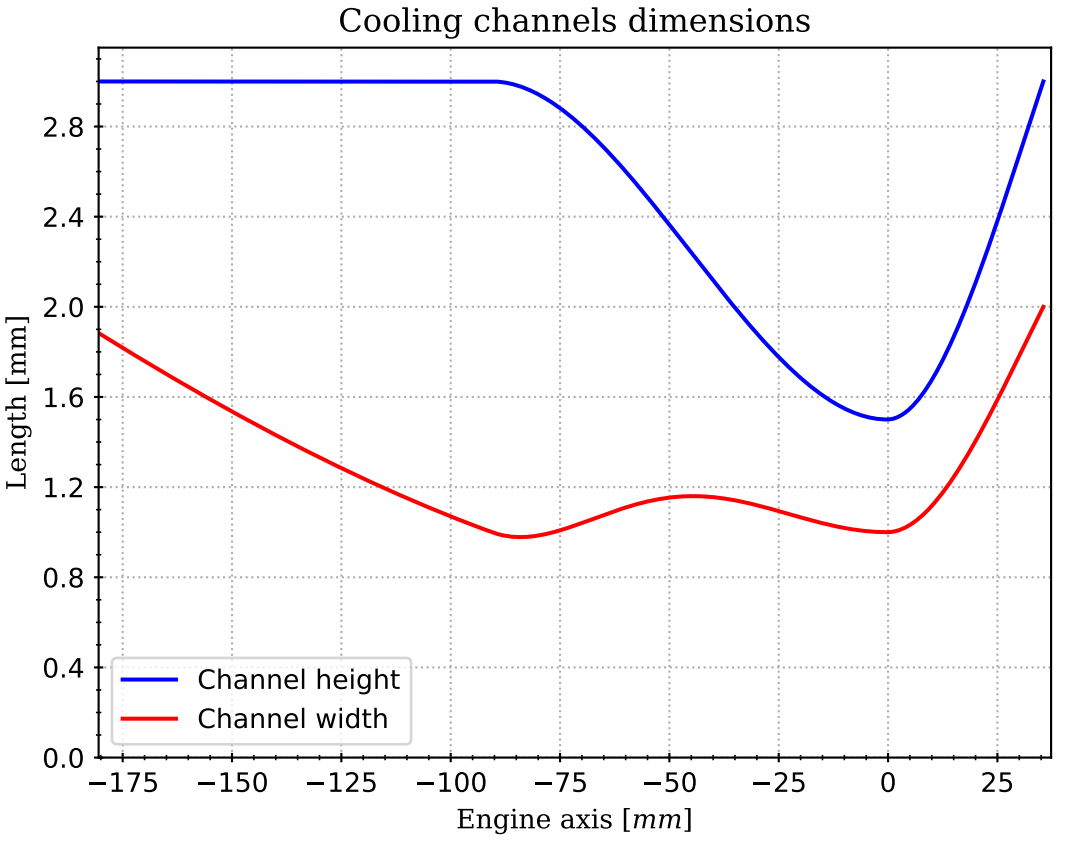
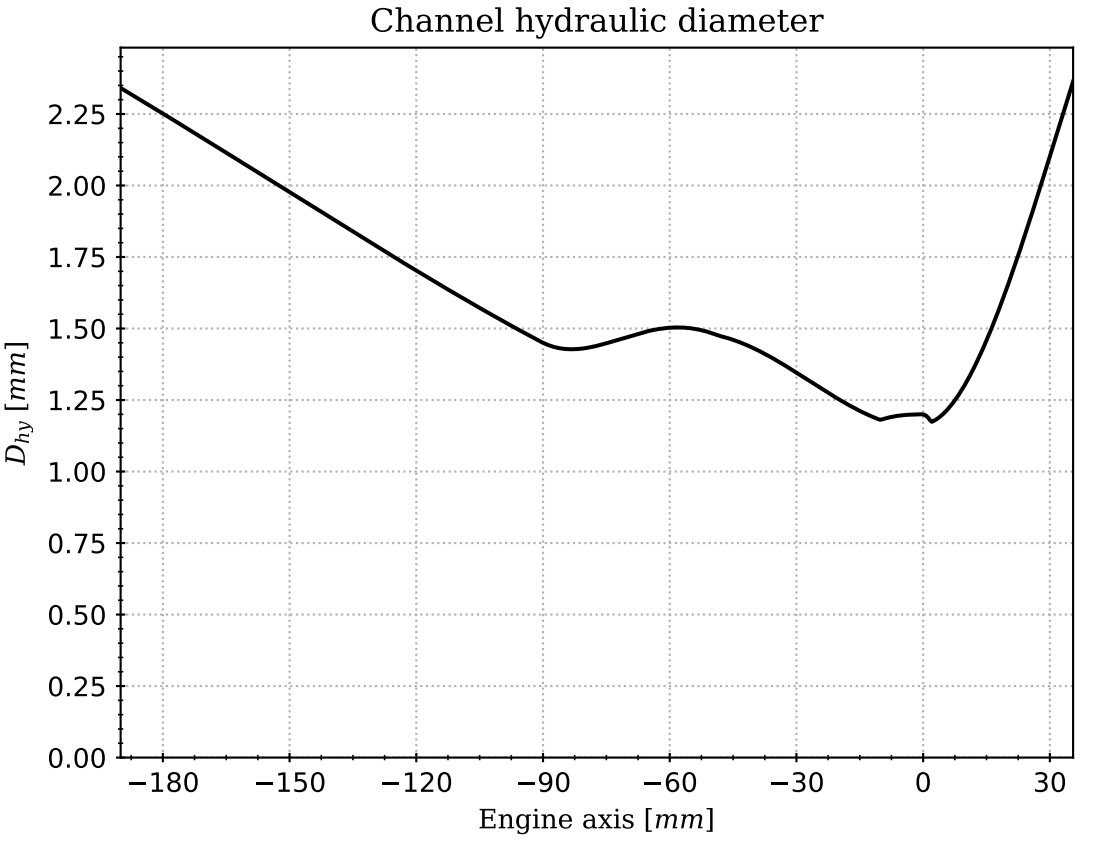

Profile of the engine
Radius [mm]
45
40
35
30
25
20
15
10
5
0
−180
−150
−120
−90
−60
−30
0
30
z-coordinate [mm]

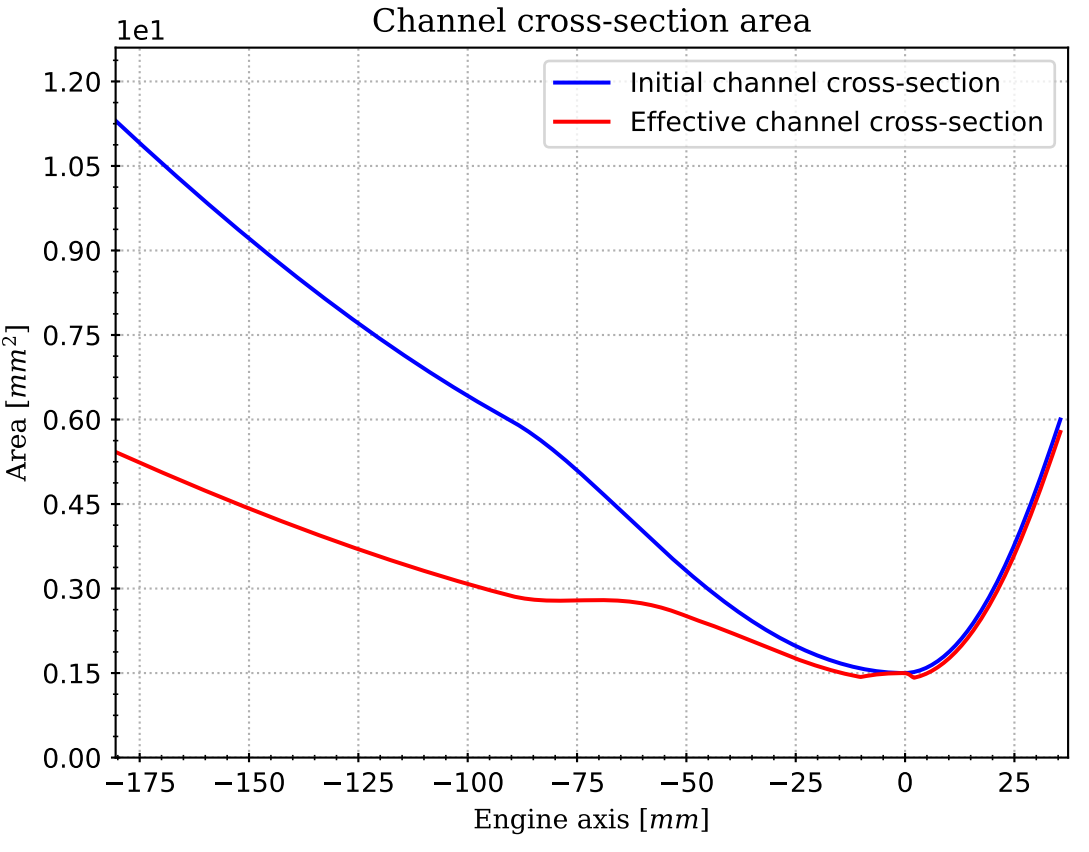

Cross-sectional area inside the engine
0.0064
0.0056
0.0048
0.0040
0.0032
0.0024
0.0016
0.0008
0.0000
Area [m^2]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

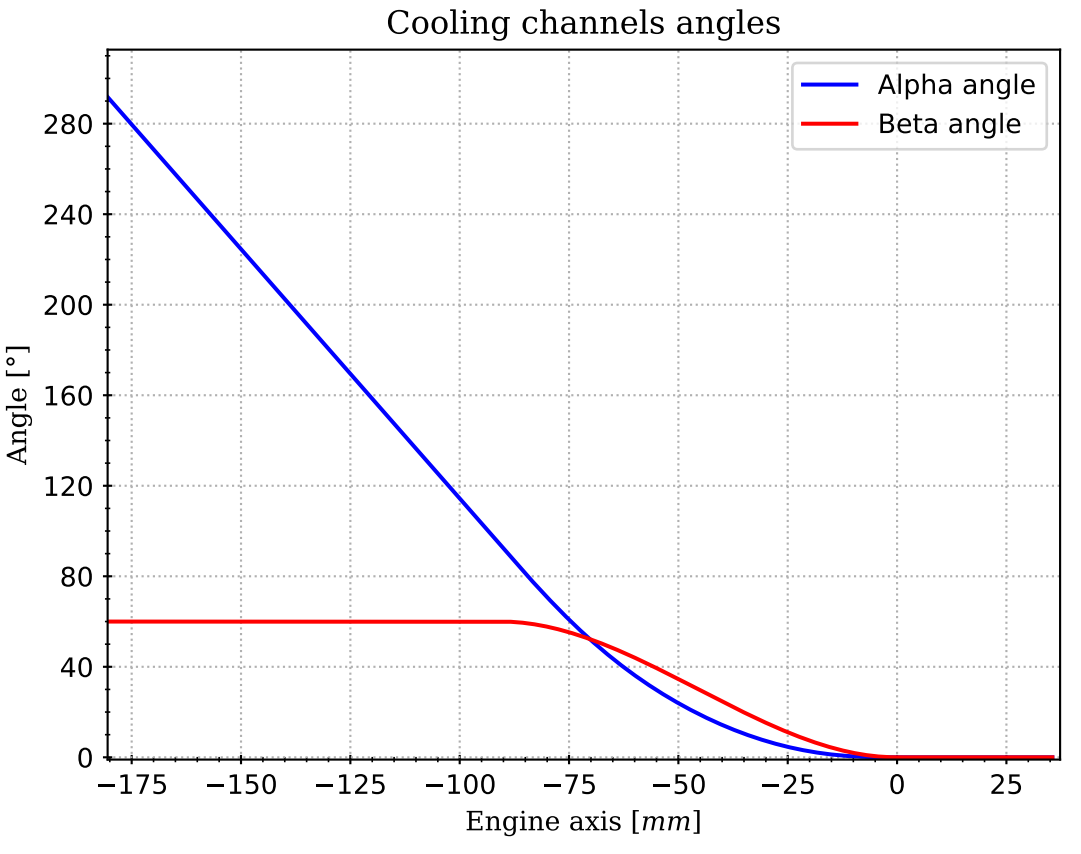

Fin dimensions
Initial fin thickness
Effective fin thickness
Length [mm]
Engine axis [mm]
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−175
−150
−125
−100
−75
−50
−25
0
25

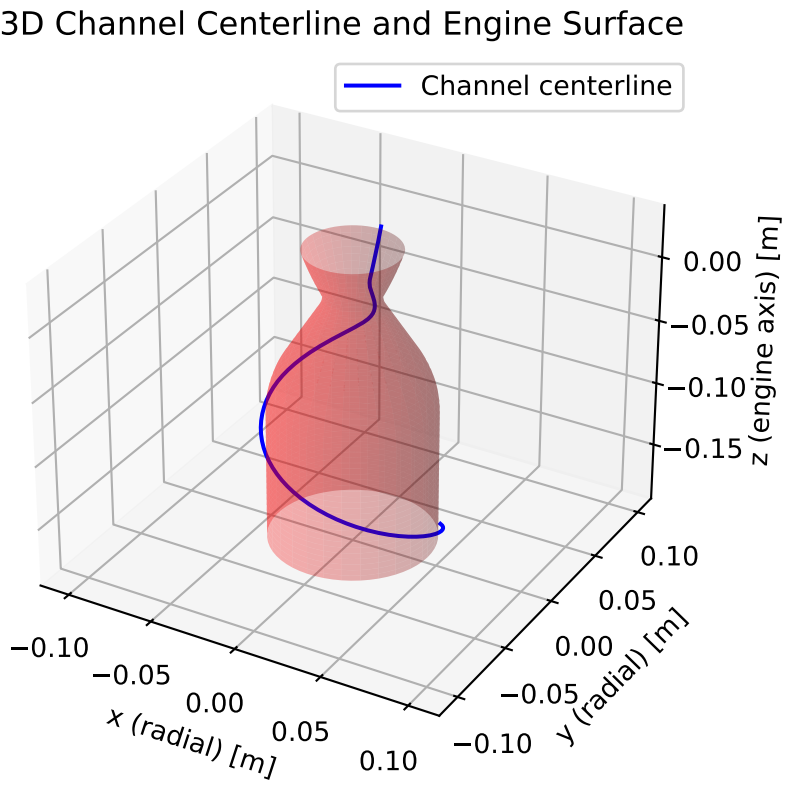

Channel aspect ratio
Aspect Ratio [-]
Engine axis [mm]
3.2
2.8
2.4
2.0
1.6
1.2
0.8
0.4
0.0
−180
−150
−120
−90
−60
−30
0
30

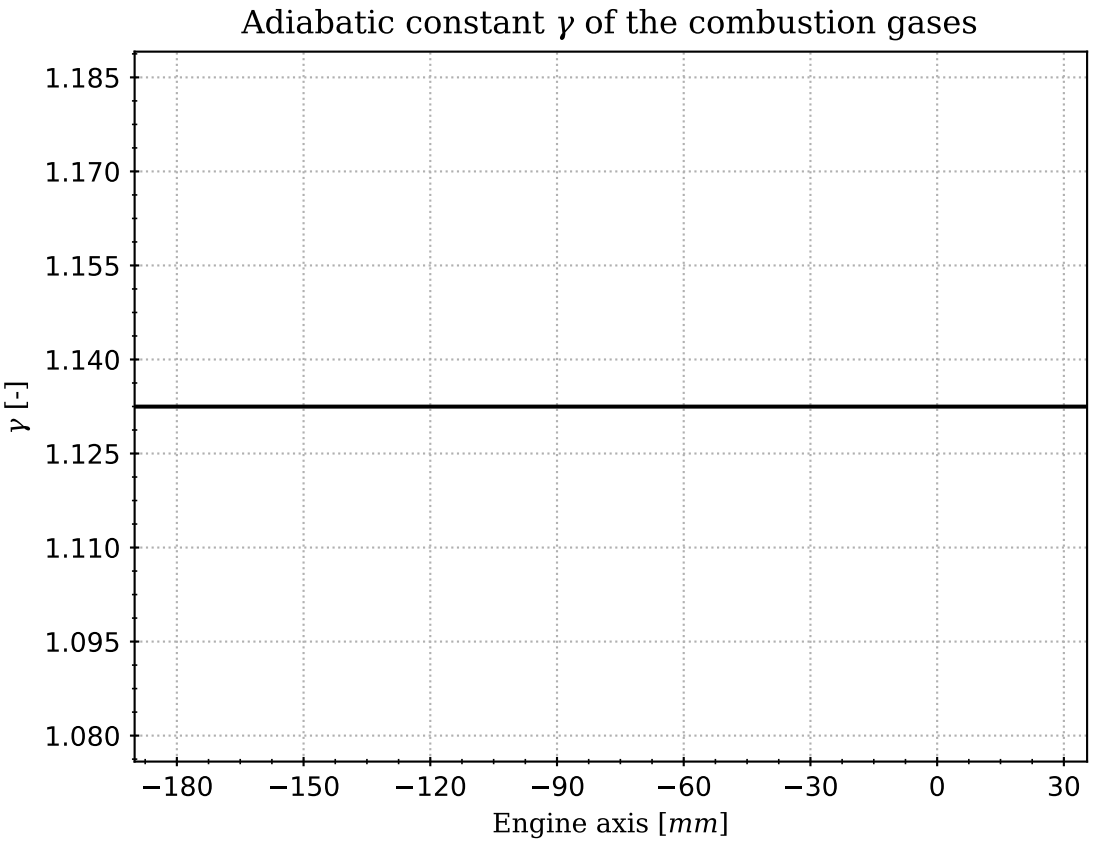

Cooling channels dimensions
Length [mm]
Engine axis [mm]
Channel height
Channel width
0.0
0.4
0.8
1.2
1.6
2.0
2.4
2.8
−175
−150
−125
−100
−75
−50
−25
0
25

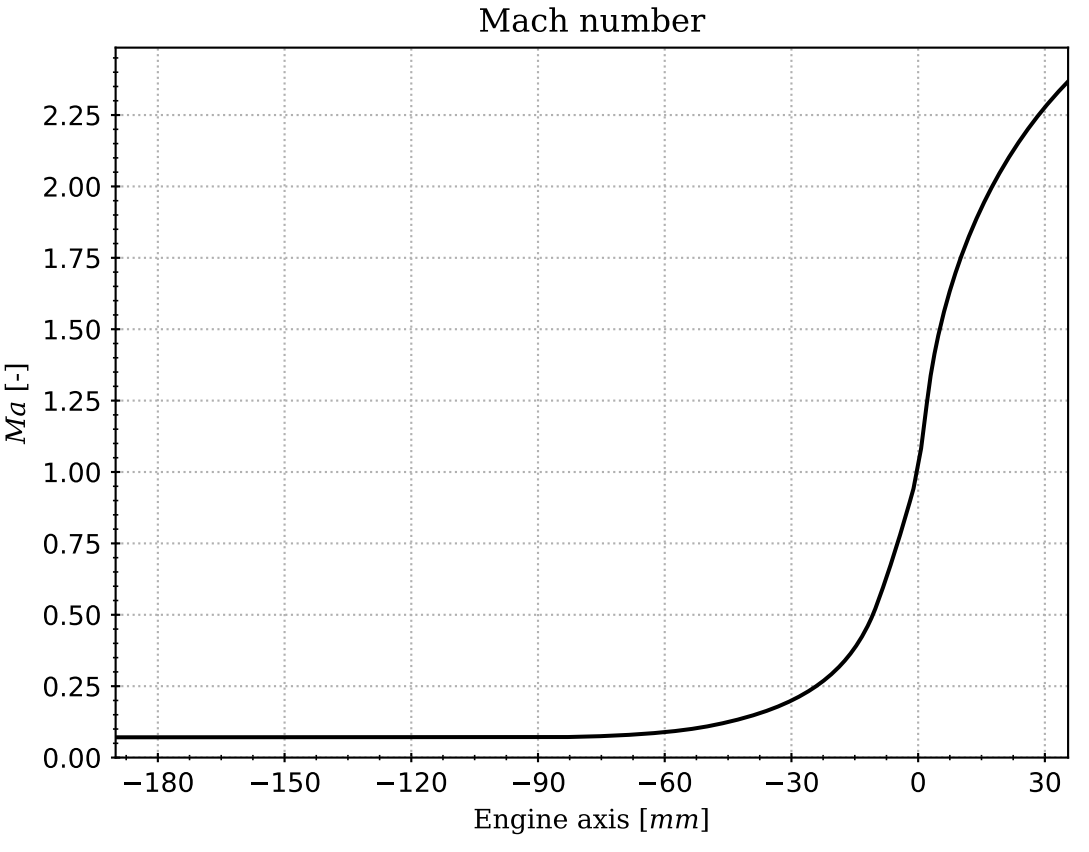

Channel hydraulic diameter
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
D_{hy} [mm]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

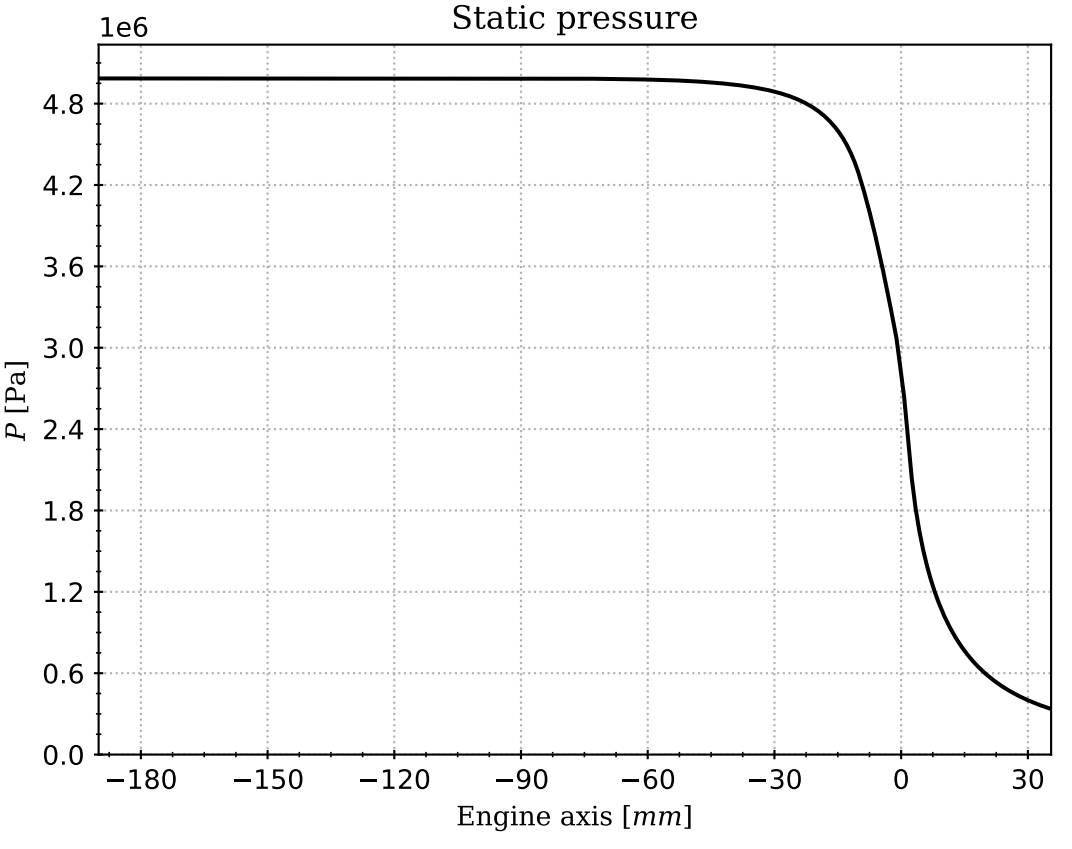

Channel cross-section area
1e1
1.20
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
Area [mm^2]
Initial channel cross-section
Effective channel cross-section
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]

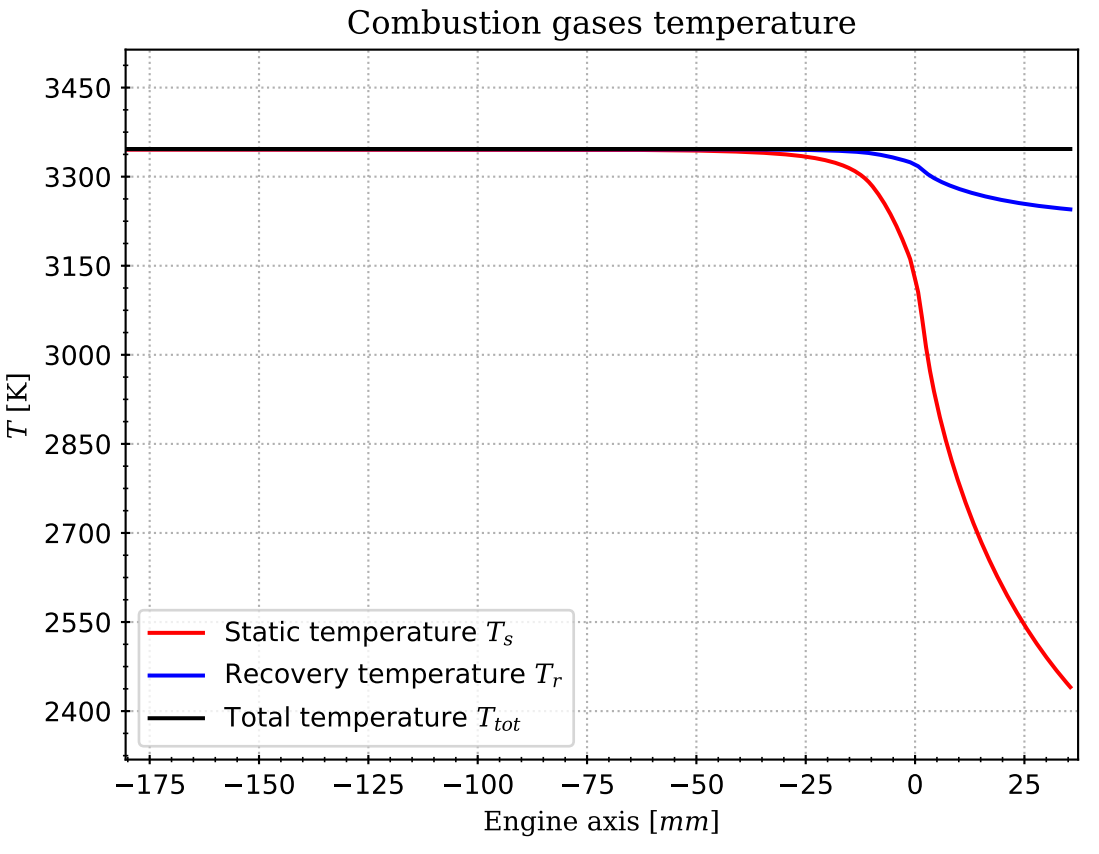

Cooling channels angles
Alpha angle
Beta angle
Angle [°]
280
240
200
160
120
80
40
0
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]

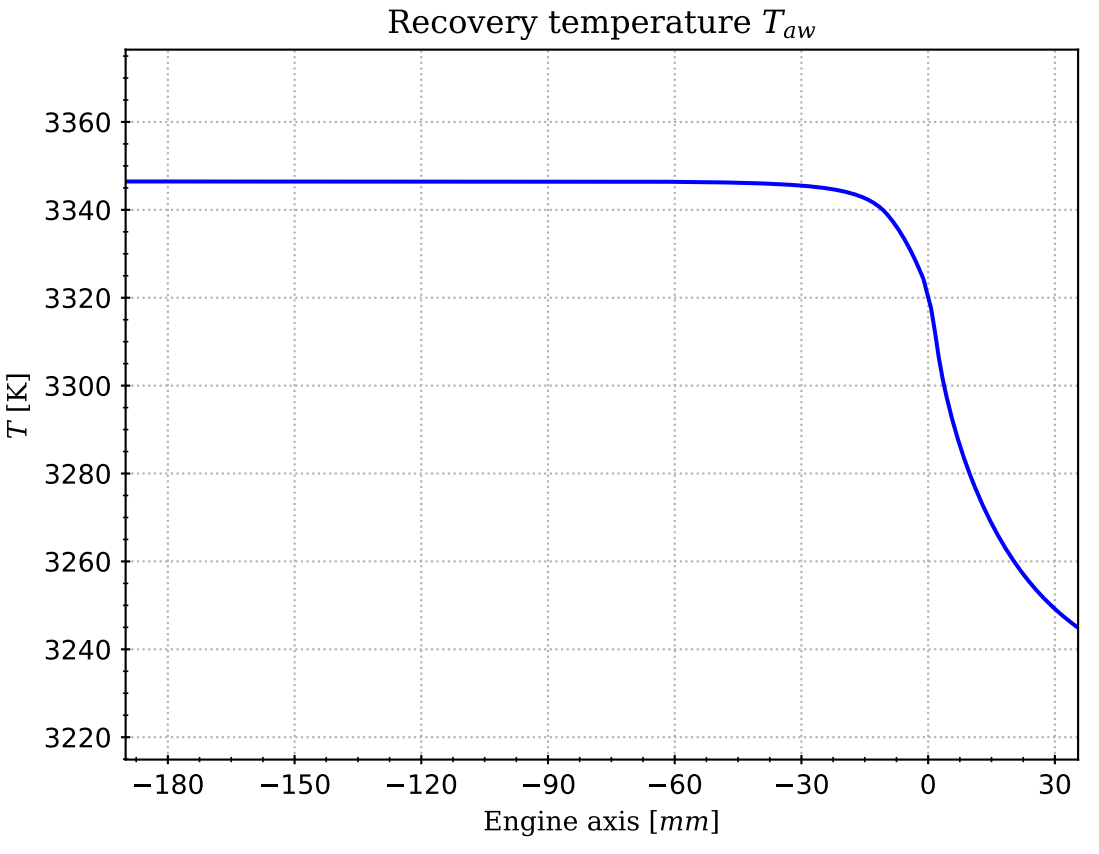

3D Channel Centerline and Engine Surface
Channel centerline
0.00
−0.05
−0.10
−0.15
z (engine axis) [m]
0.10
0.05
0.00
−0.05
−0.10
y (radial) [m]
−0.10
−0.05
0.00
0.05
0.10
x (radial) [m]

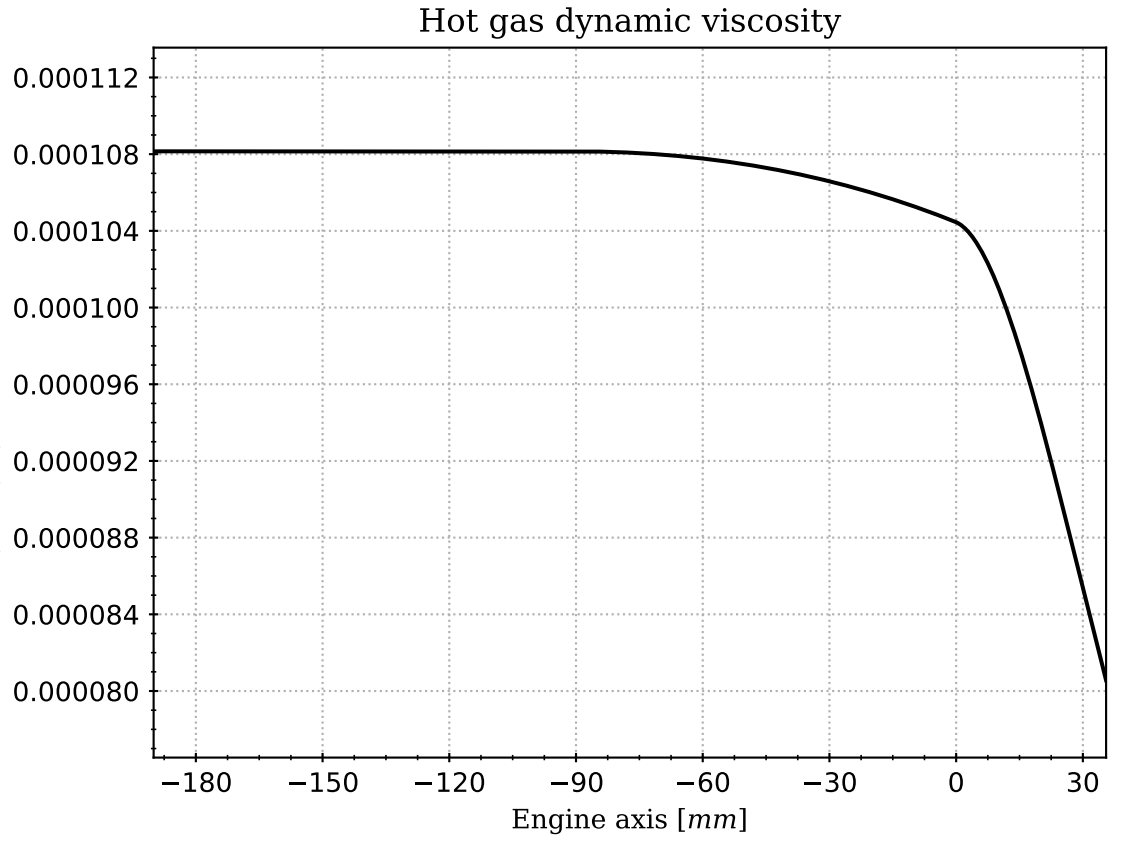

Adiabatic constant γ of the combustion gases
1.185
1.170
1.155
1.140
1.125
1.110
1.095
1.080
γ [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

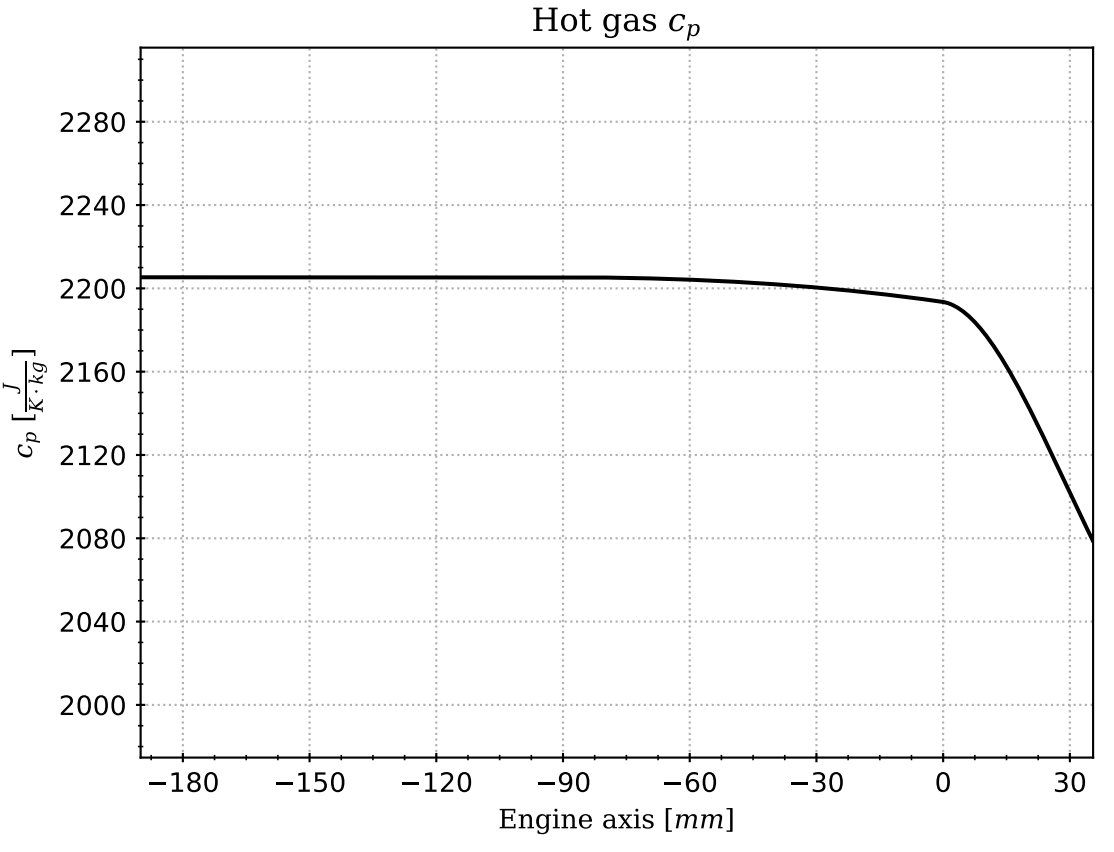

Mach number
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Ma [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

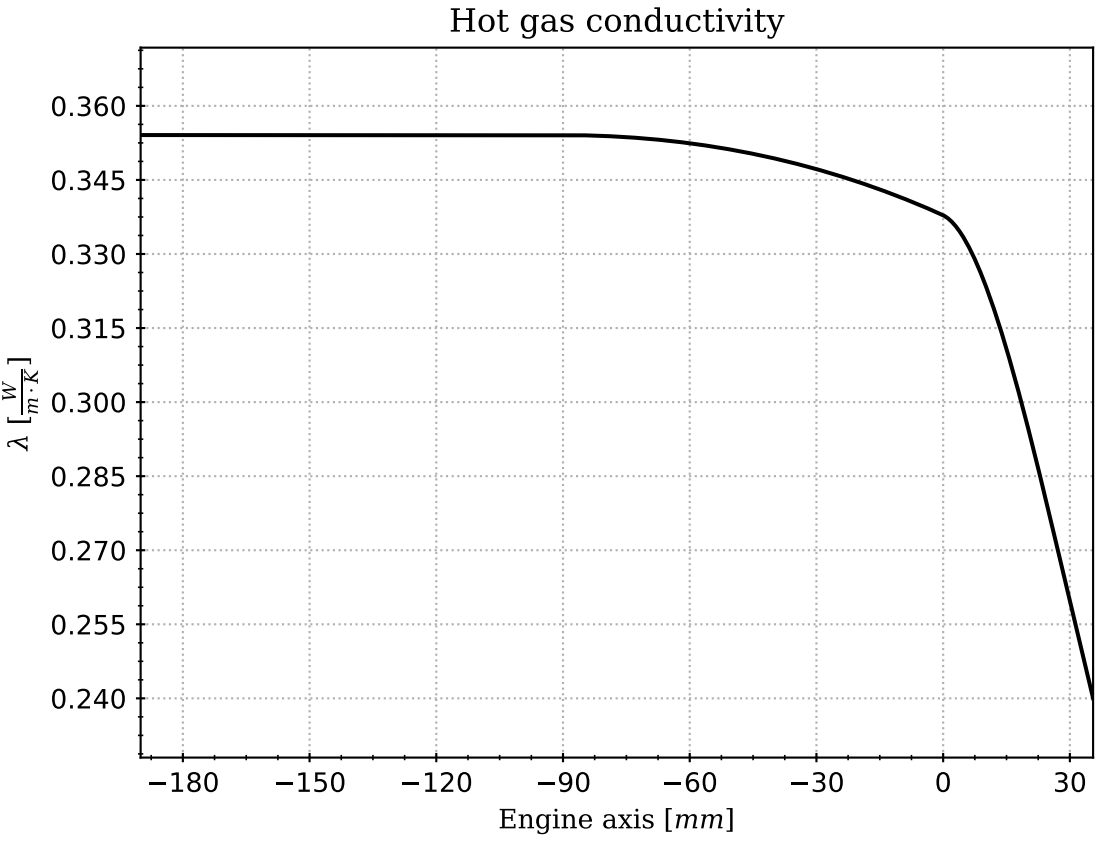

Static pressure
1e6
4.8
4.2
3.6
3.0
2.4
1.8
1.2
0.6
0.0
P [Pa]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

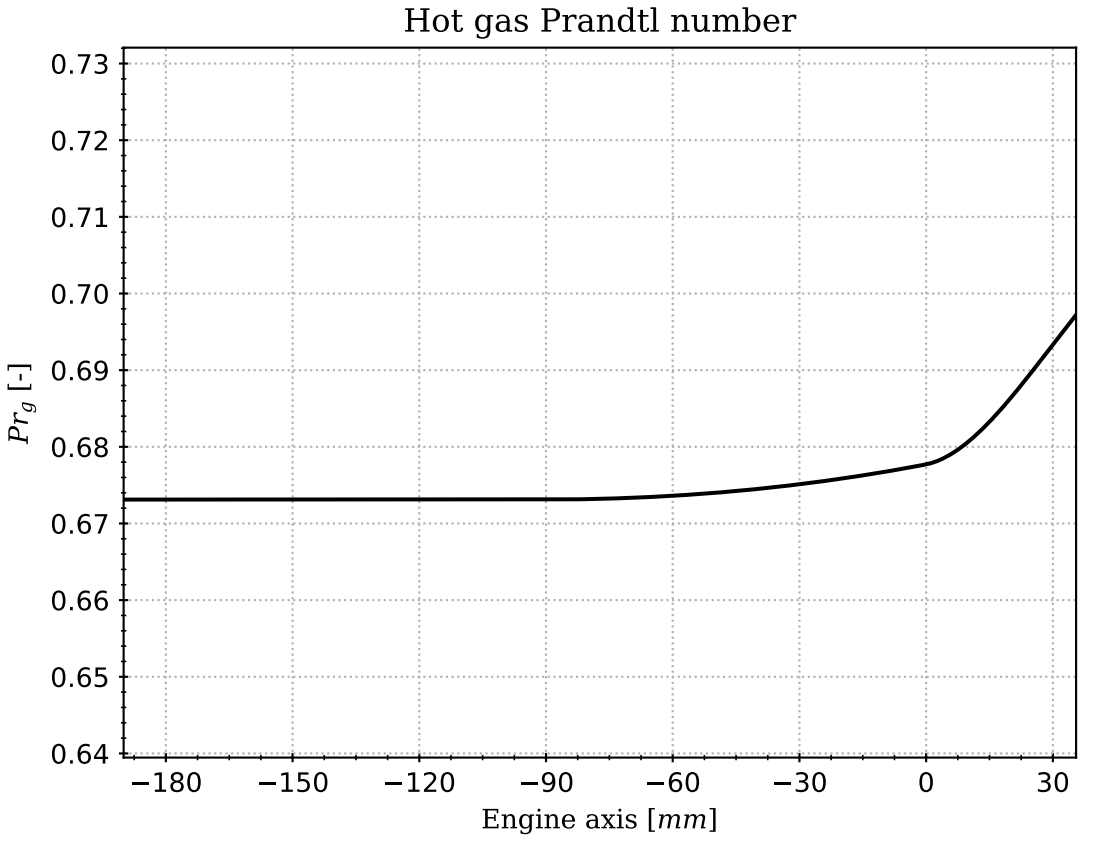

Combustion gases temperature
3450
3300
3150
3000
2850
2700
2550
2400
T [K]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]
Static temperature T_s
Recovery temperature T_r
Total temperature T_{tot}

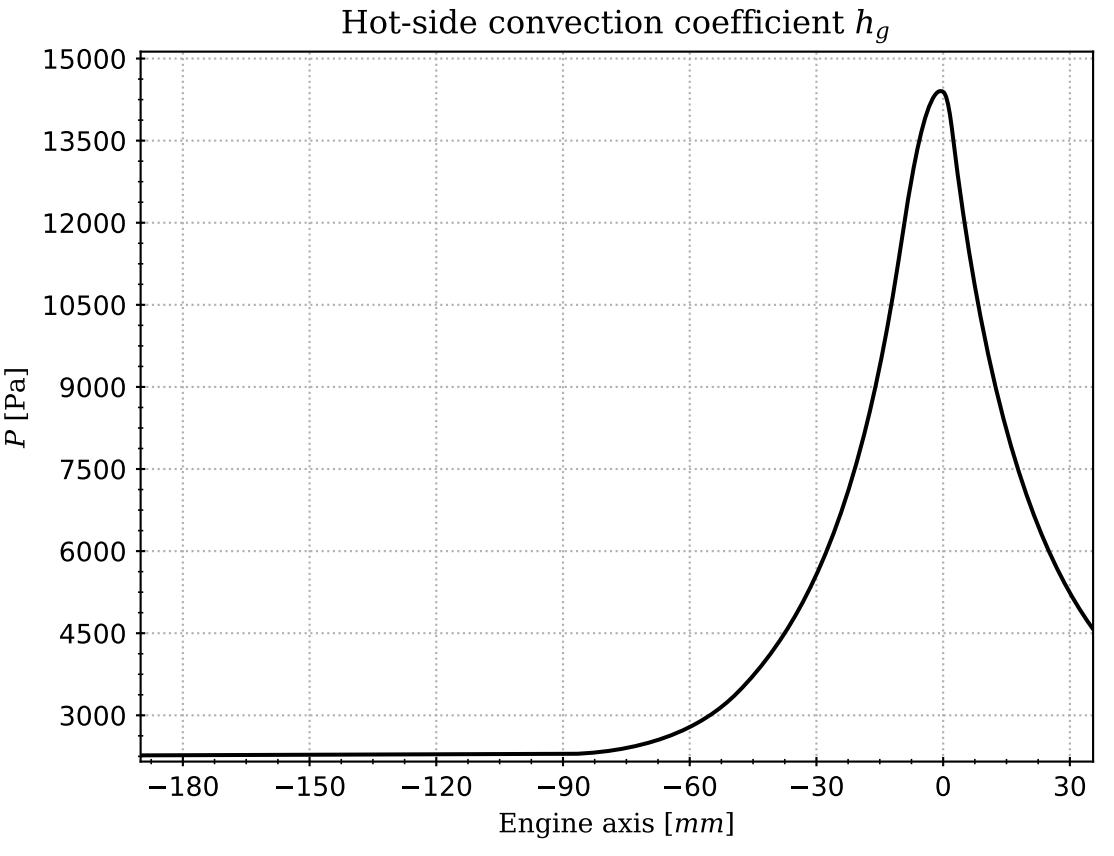

Recovery temperature T_{aw}
3360
3340
3320
3300
3280
3260
3240
3220
T [K]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

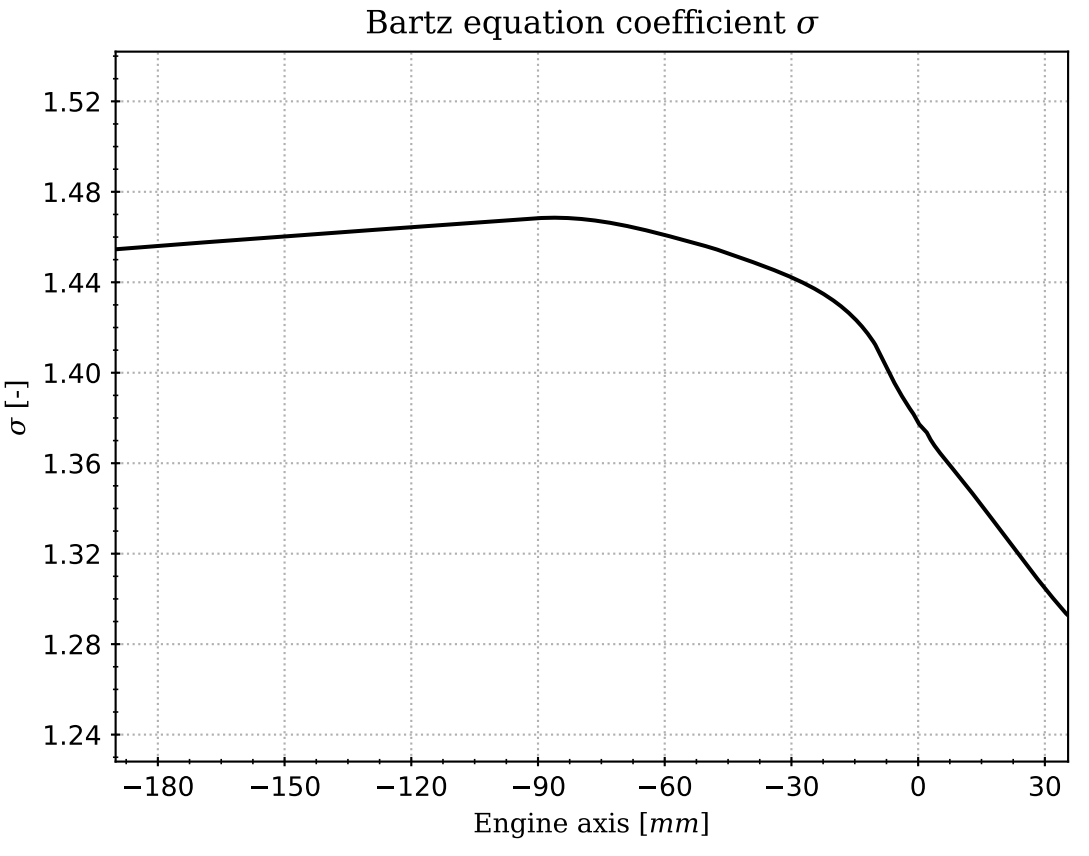

Hot gas dynamic viscosity
0.000112
0.000108
0.000104
0.000100
0.000096
0.000092
0.000088
0.000084
0.000080
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

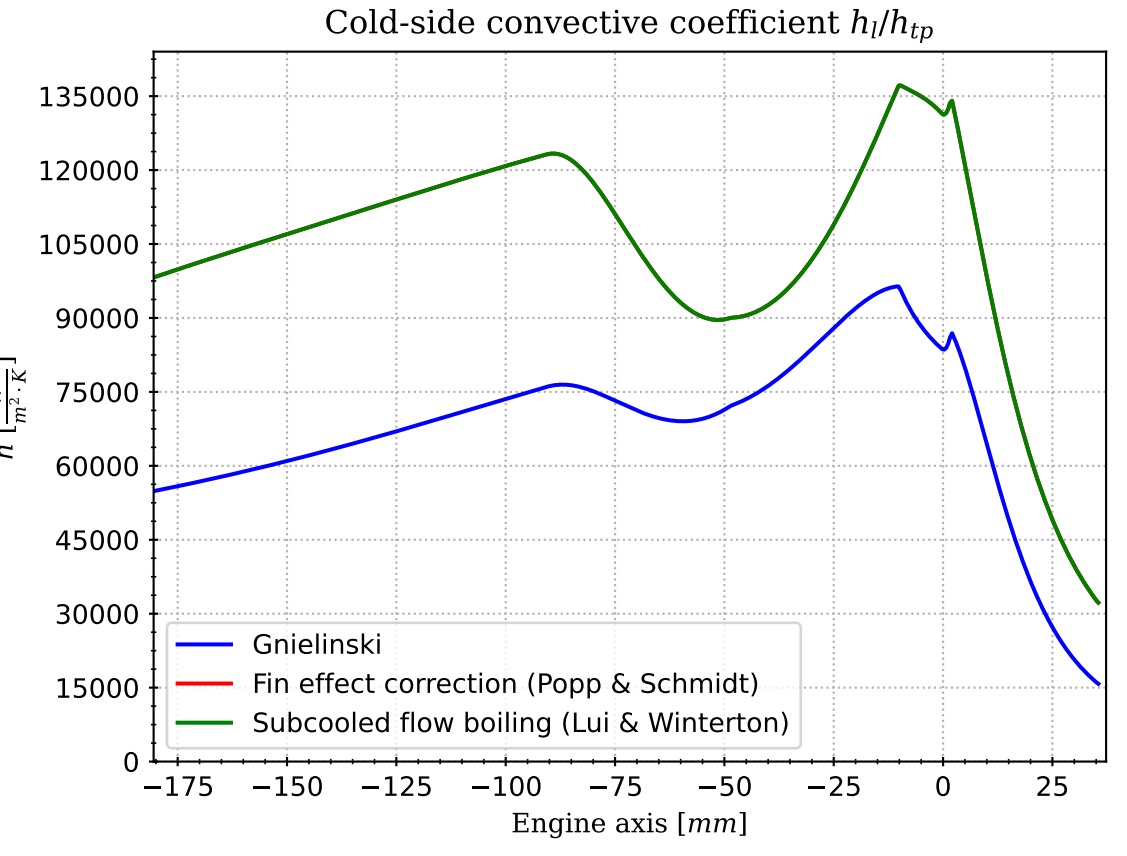

Hot gas c_p
2280
2240
2200
2160
2120
2080
2040
2000
c_p $[\frac{J}{K \cdot kg}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$

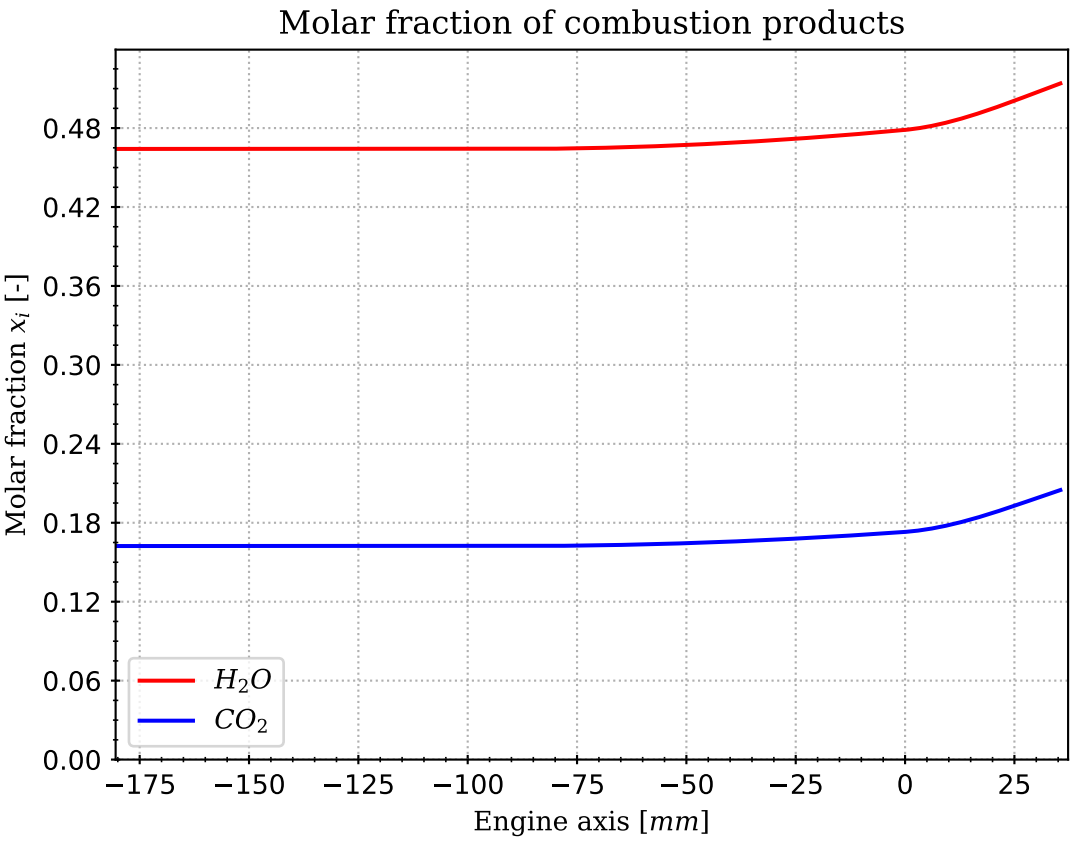

Hot gas conductivity
0.360
0.345
0.330
0.315
0.300
0.285
0.270
0.255
0.240
$\lambda\ [\frac{W}{m \cdot K}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

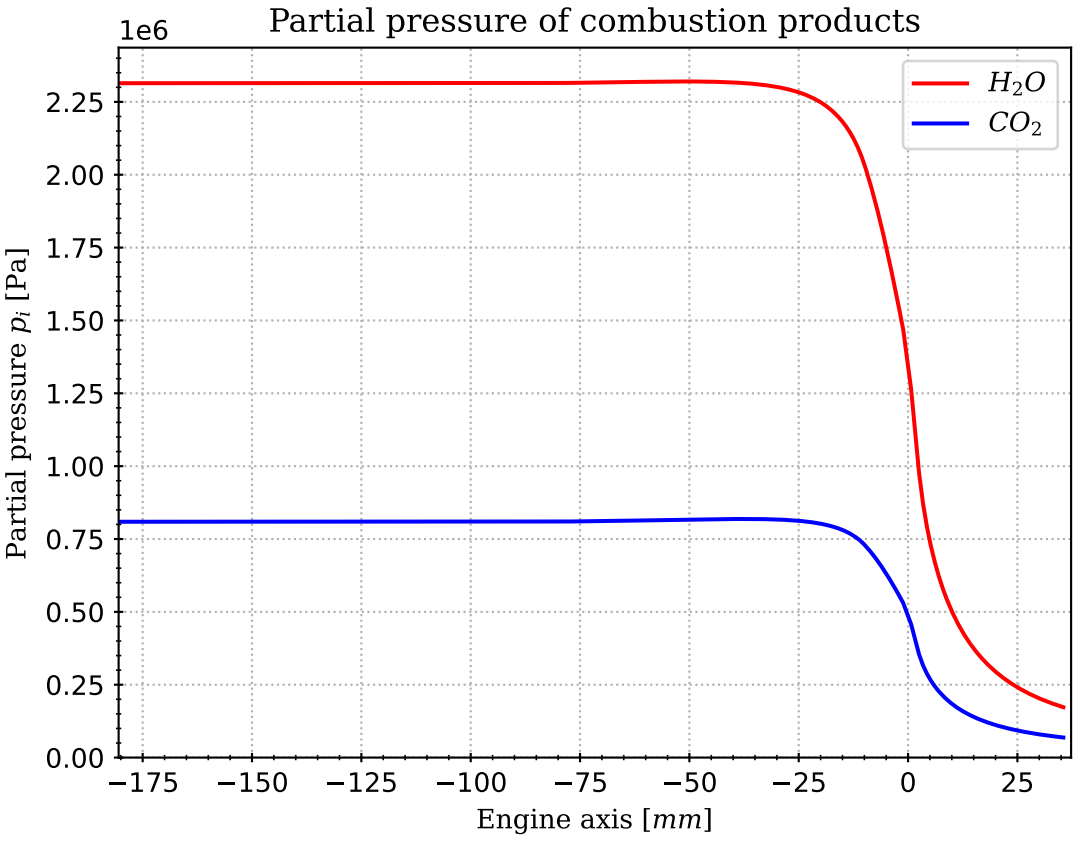

Hot gas Prandtl number
0.73
0.72
0.71
0.70
0.69
0.68
0.67
0.66
0.65
0.64
Pr_g [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

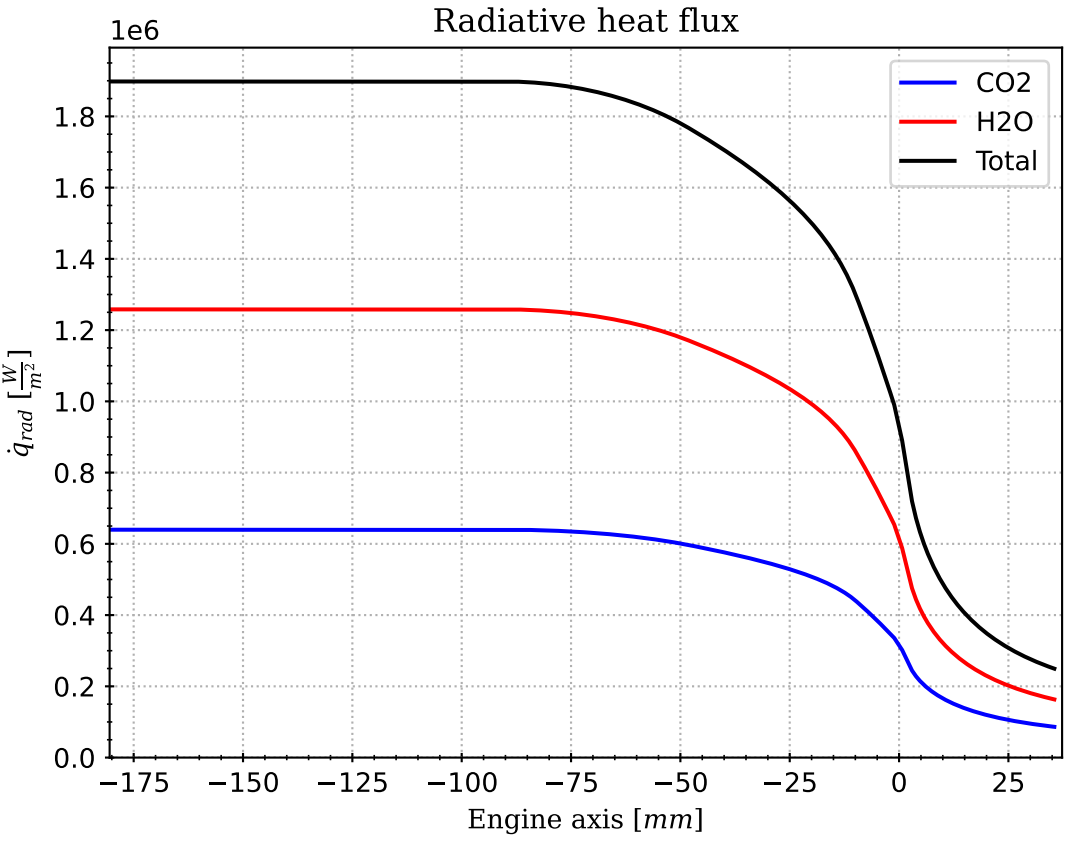

Hot-side convection coefficient h_g
15000
13500
12000
10500
9000
7500
6000
4500
3000
P [Pa]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

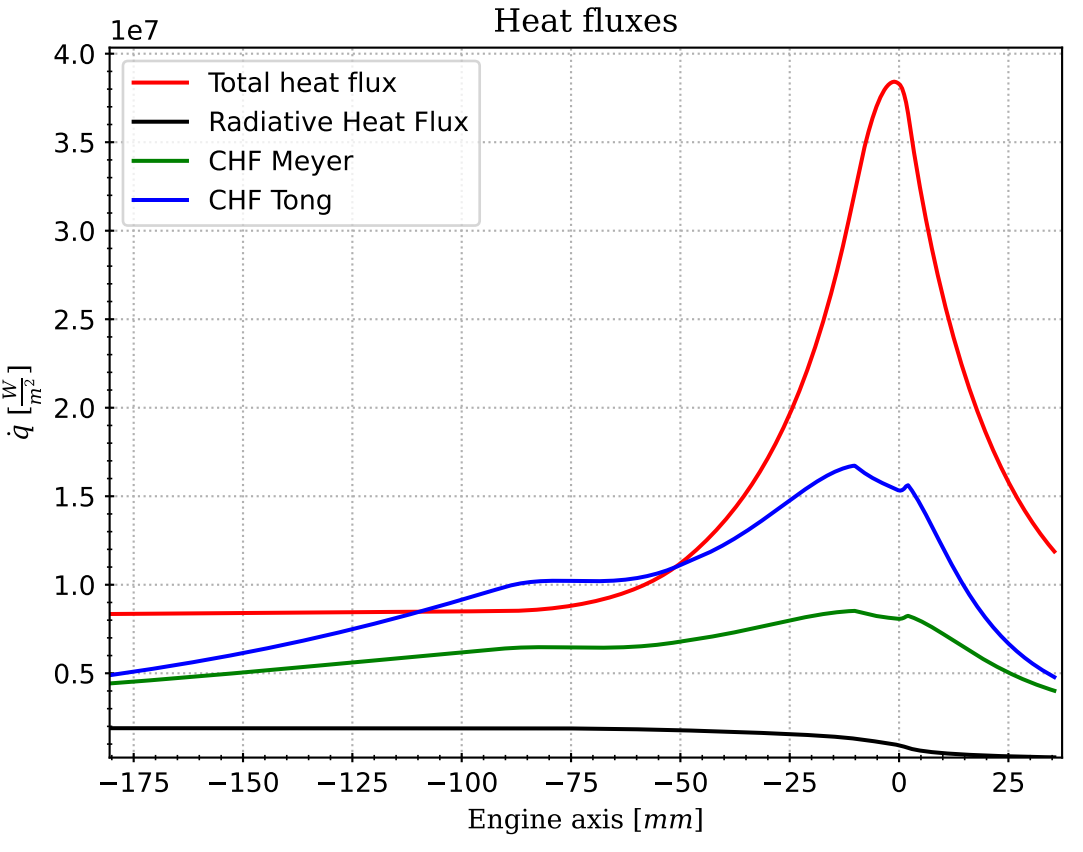

Bartz equation coefficient σ
1.52
1.48
1.44
1.40
1.36
1.32
1.28
1.24
σ [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

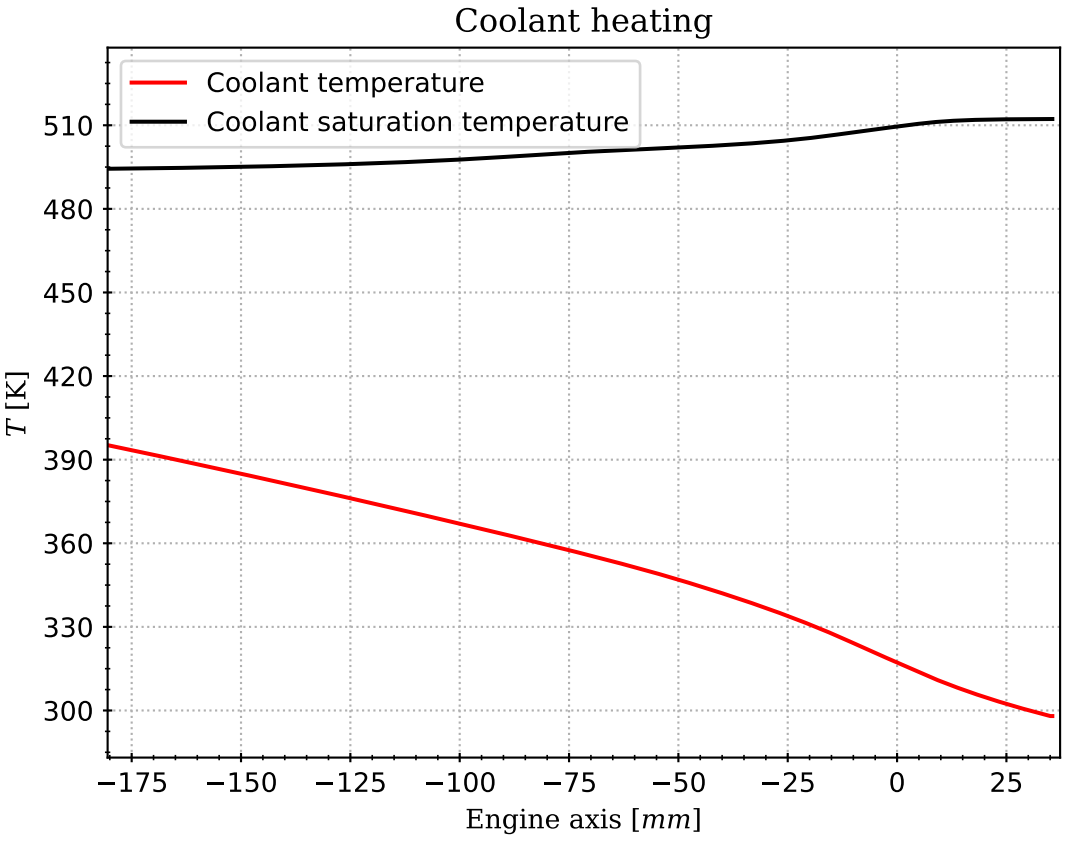

Cold-side convective coefficient h_l/h_{tp}
135000
120000
105000
90000
75000
60000
45000
30000
15000
0
$h\ [\frac{W}{m^2 \cdot K}]$
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis $[mm]$
Gnielinski
Fin effect correction (Popp & Schmidt)
Subcooled flow boiling (Lui & Winterton)

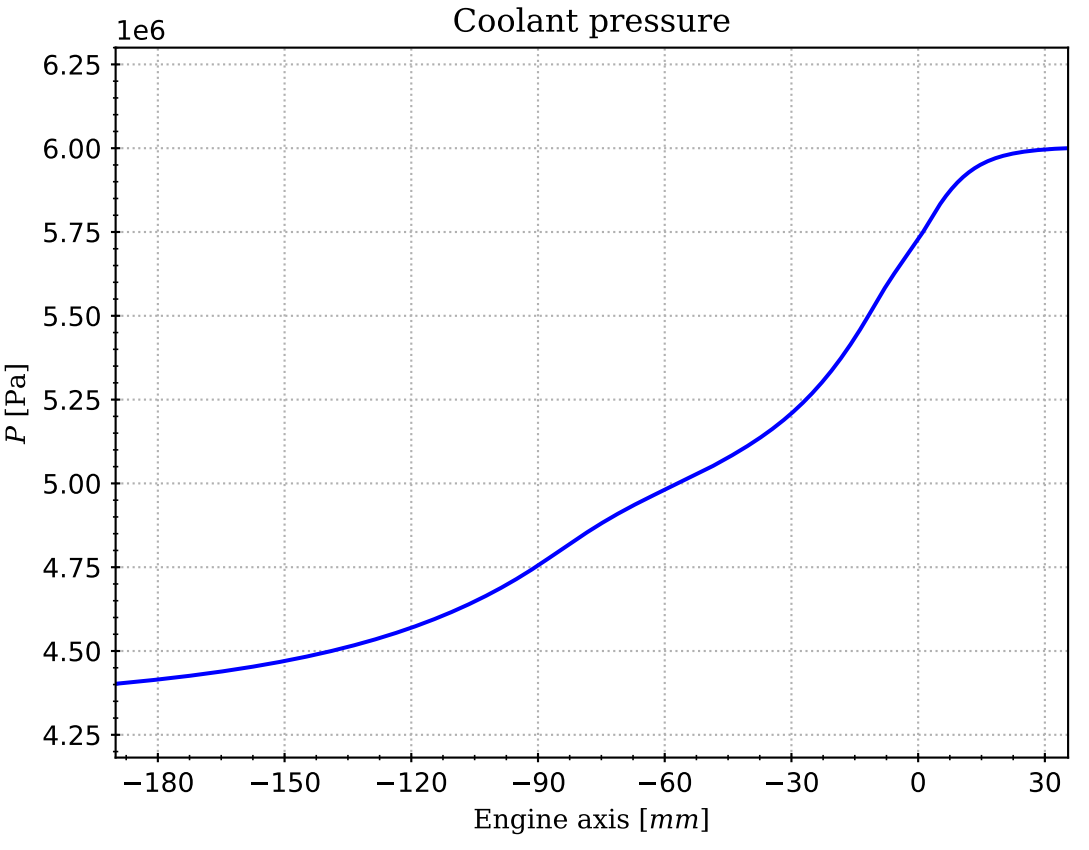

Molar fraction of combustion products
Molar fraction x_i [-]
0.48
0.42
0.36
0.30
0.24
0.18
0.12
0.06
0.00
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]
H_2O
CO_2

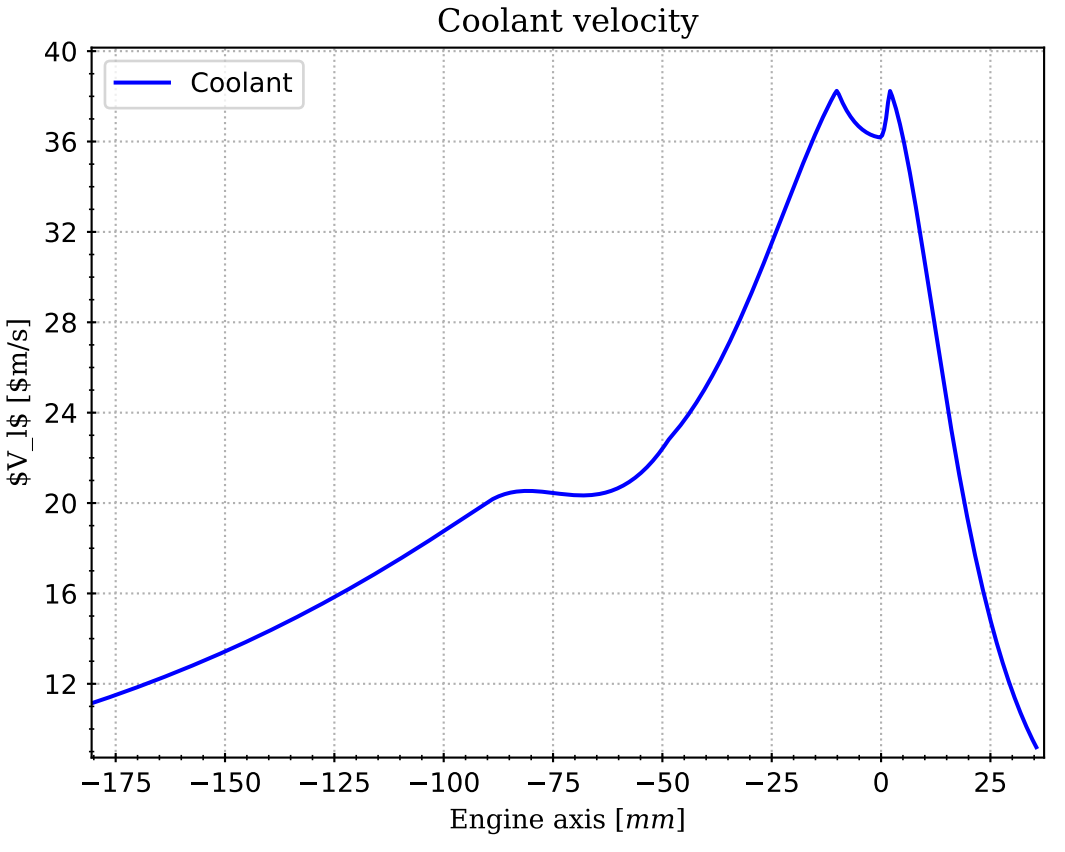

Partial pressure of combustion products
1e6
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Partial pressure p_i [Pa]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]
H_2O
CO_2

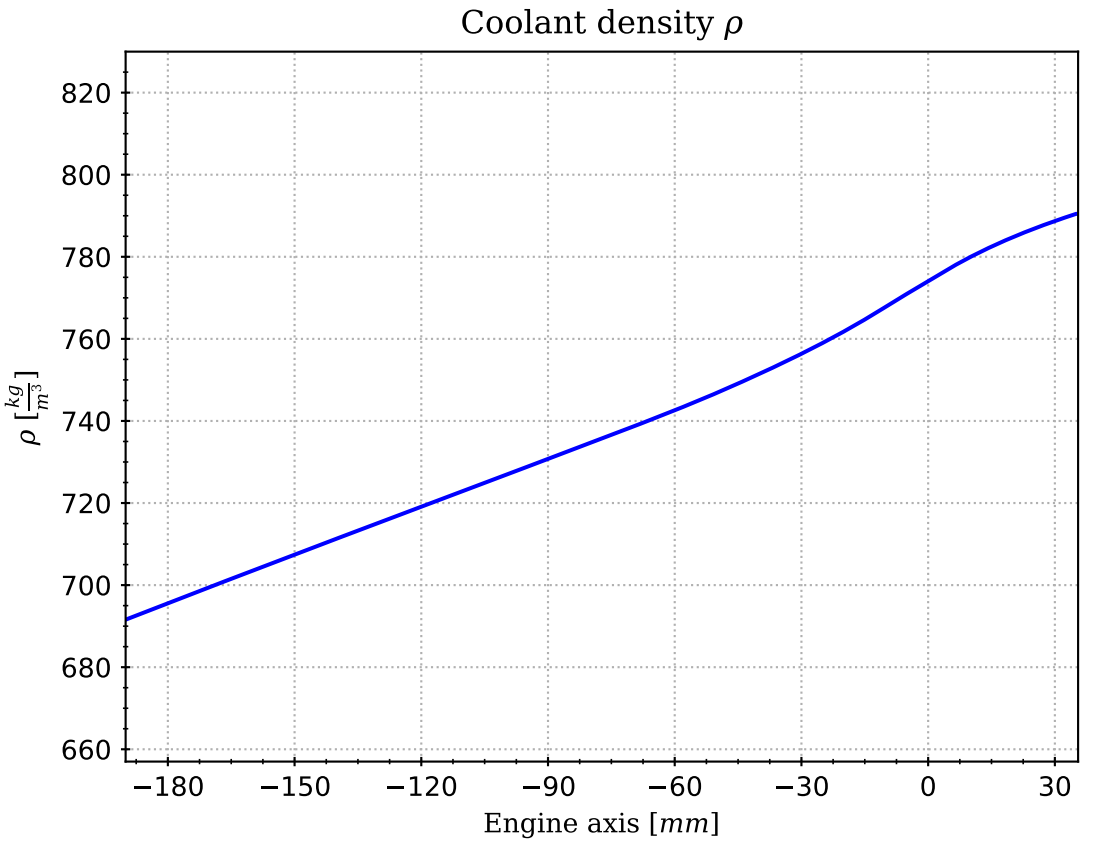

Radiative heat flux
1e6
CO2
H2O
Total
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
$\dot{q}_{rad}$ $[\frac{W}{m^2}]$
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis $[mm]$

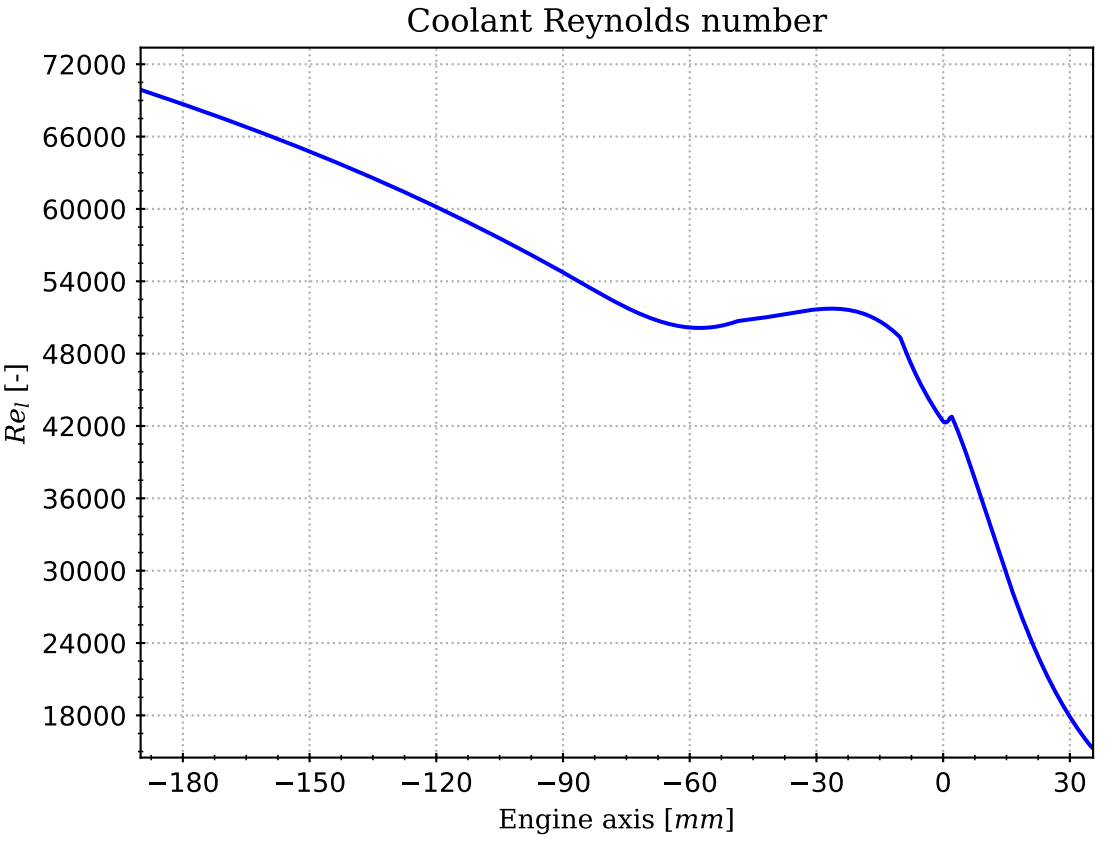

Heat fluxes
1e7
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
$\dot{q}$ [$\frac{W}{m^2}$]
Total heat flux
Radiative Heat Flux
CHF Meyer
CHF Tong
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]

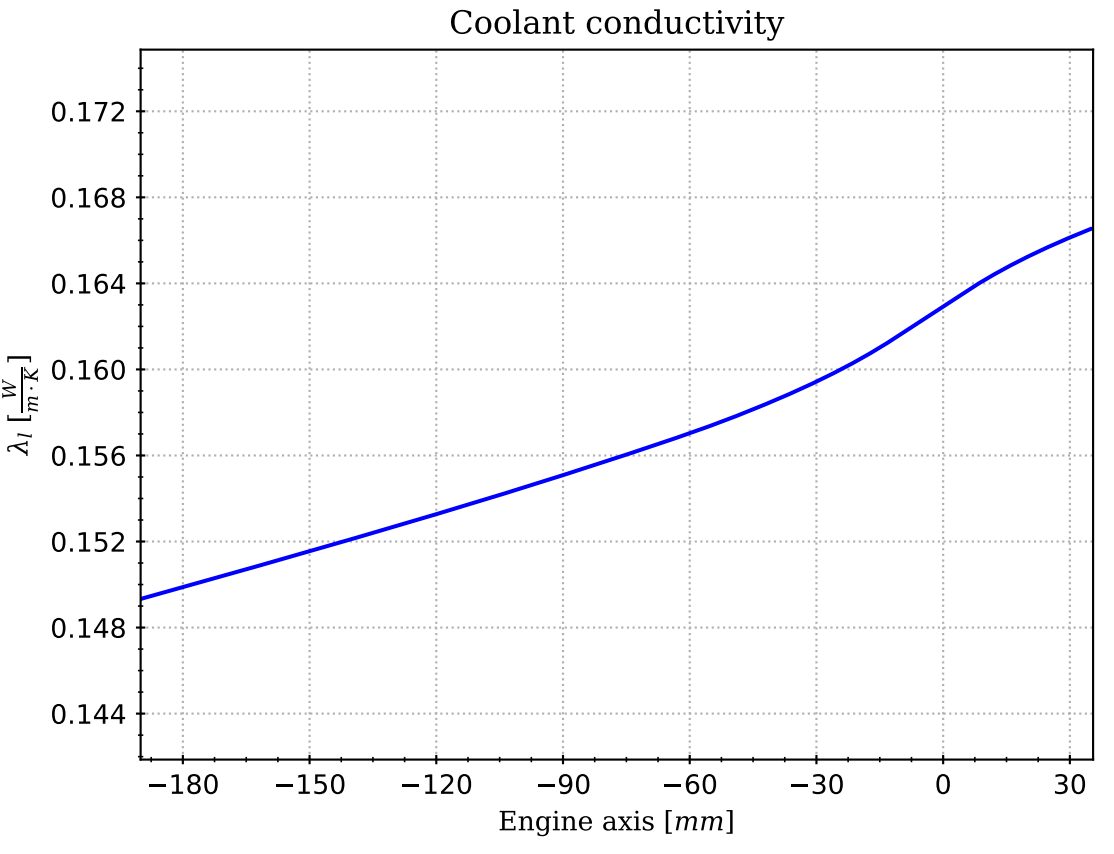

Coolant heating
Coolant temperature
Coolant saturation temperature
510
480
450
420
390
360
330
300
T [K]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]

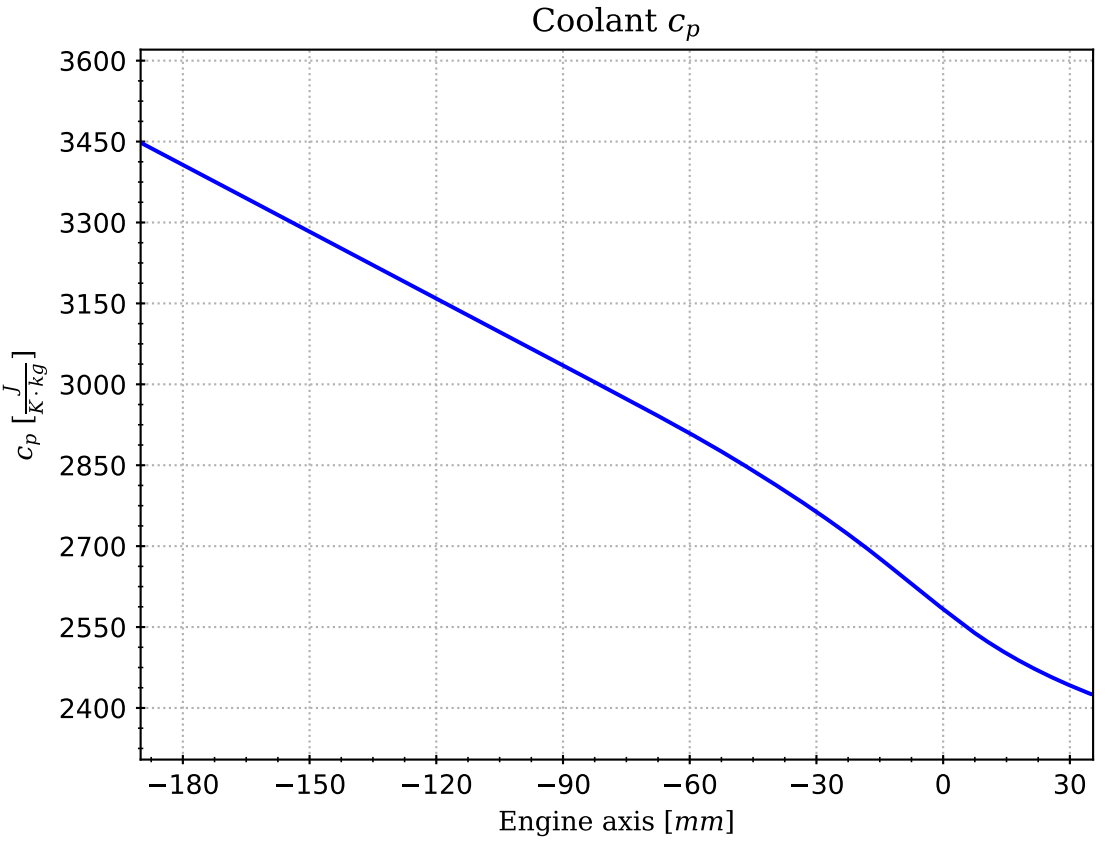

Coolant pressure
1e6
6.25
6.00
5.75
5.50
5.25
5.00
4.75
4.50
4.25
P [Pa]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

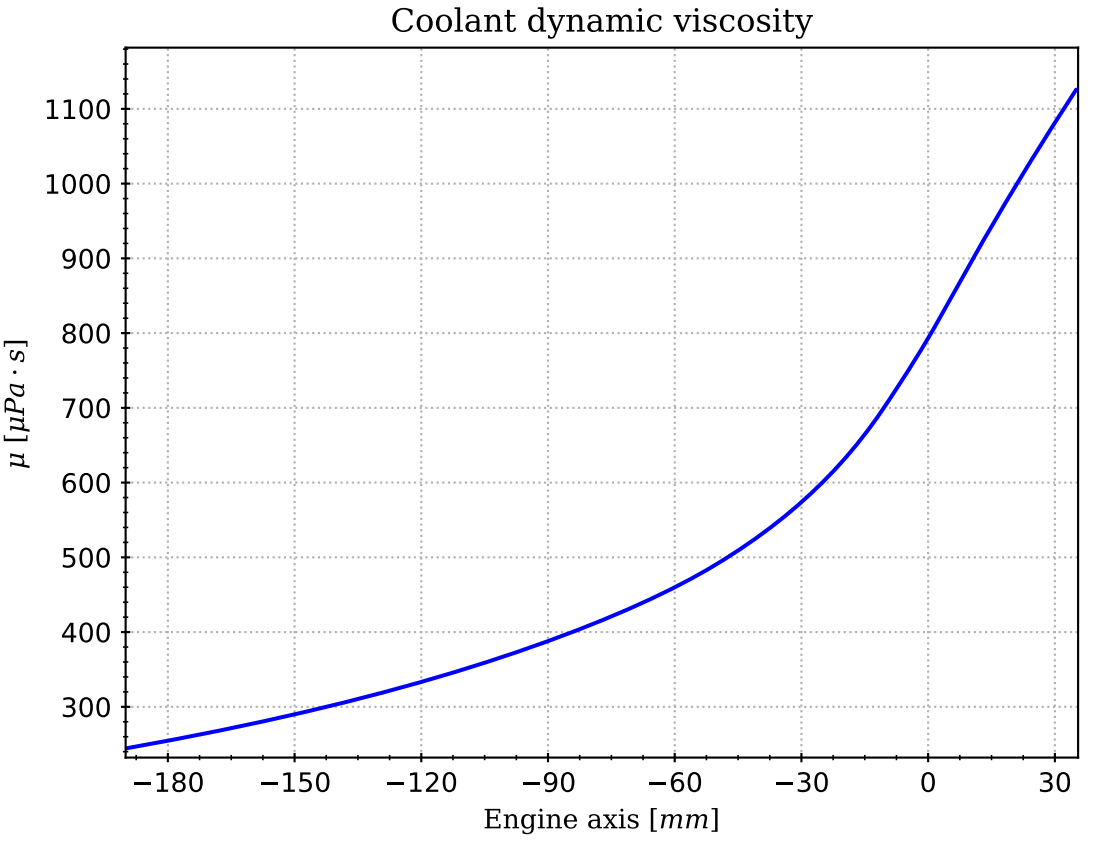

Coolant velocity
Coolant
40
36
32
28
24
20
16
12
V_l [$m/s]
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]

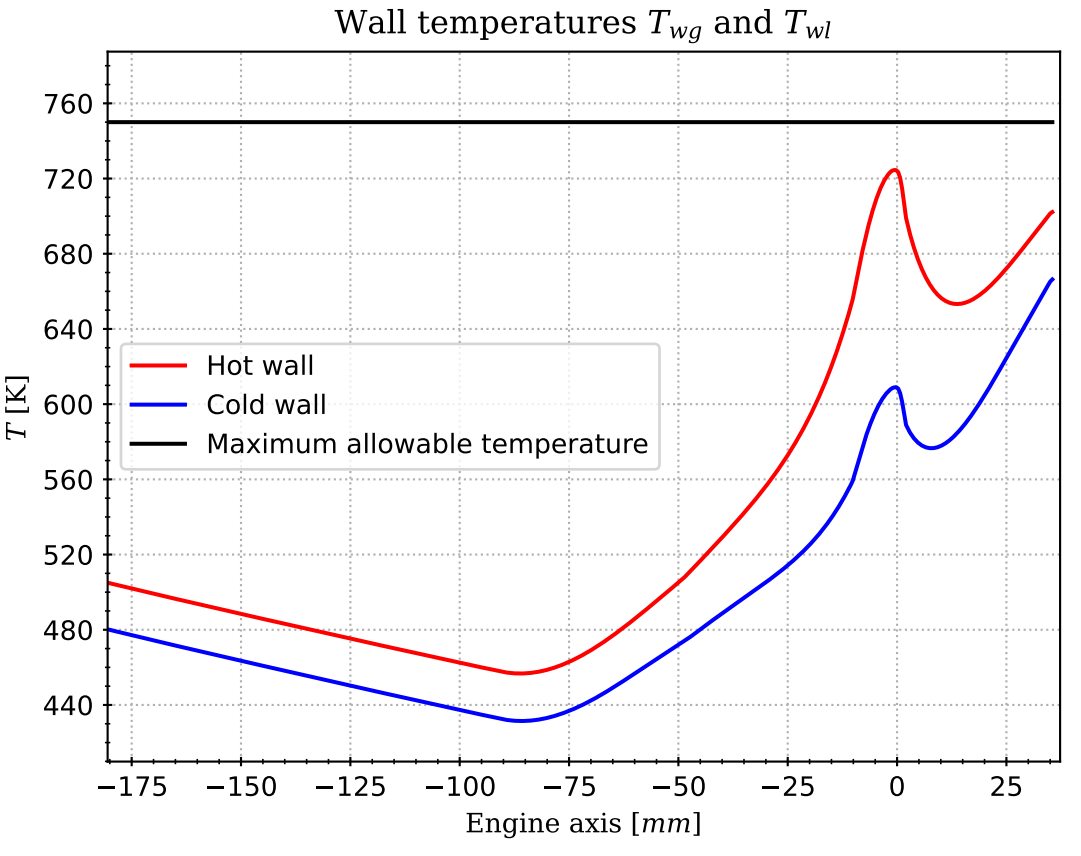

Coolant density ρ
820
800
780
760
740
720
700
680
660
$\rho\ [\frac{kg}{m^3}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$

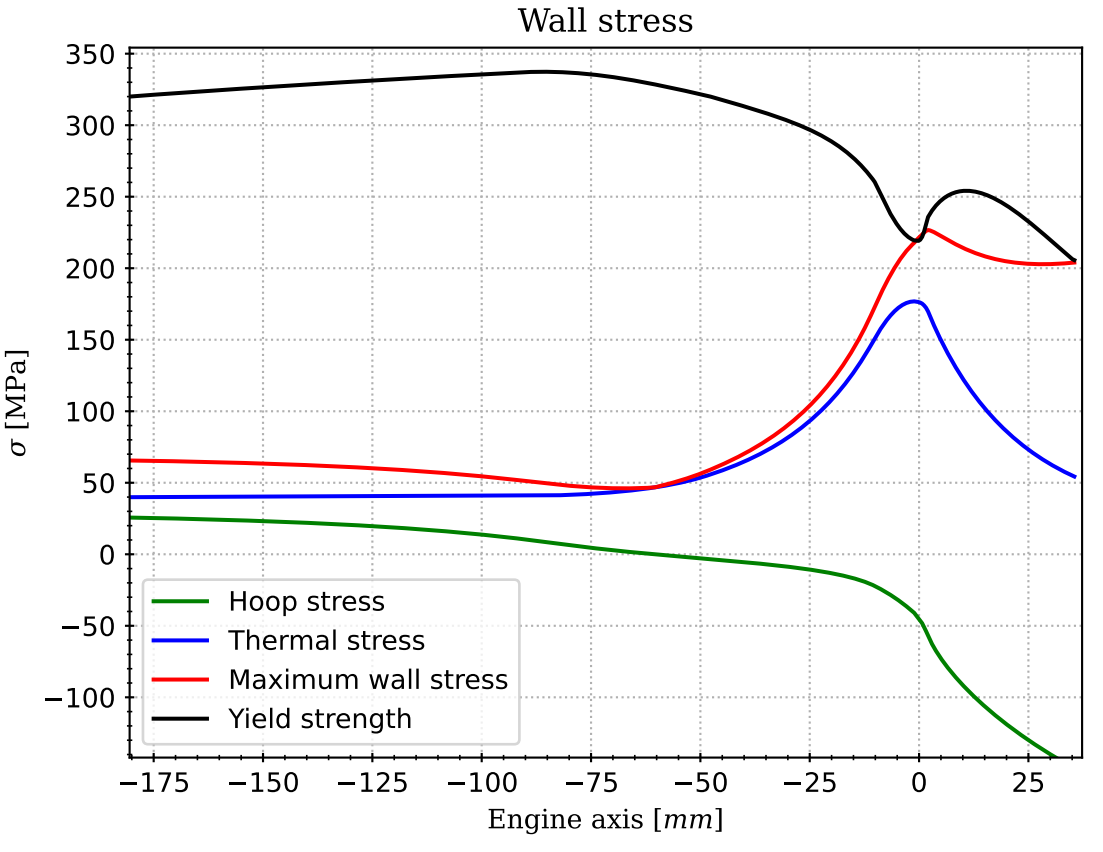

Coolant Reynolds number
72000
66000
60000
54000
48000
42000
36000
30000
24000
18000
Re_l [-]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]

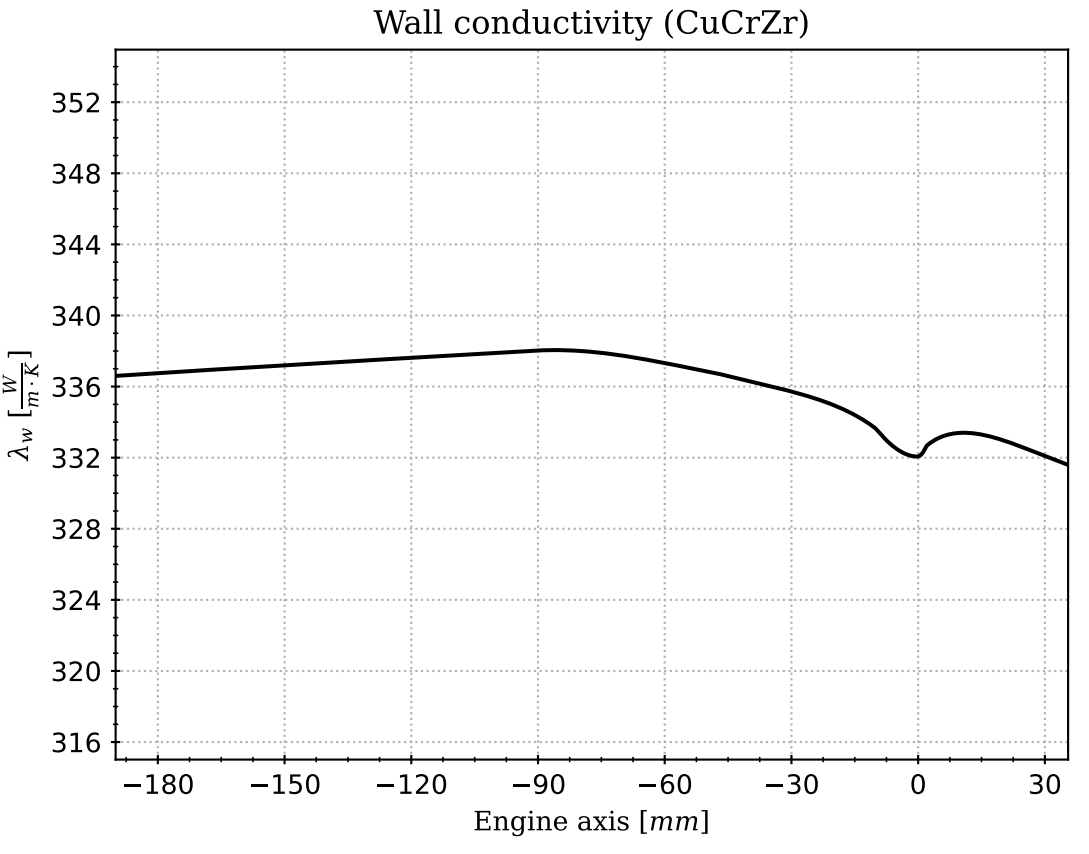

Coolant conductivity
0.172
0.168
0.164
0.160
0.156
0.152
0.148
0.144
λ_l $[\frac{W}{m \cdot K}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$


Coolant c_p
3600
3450
3300
3150
3000
2850
2700
2550
2400
c_p $[\frac{J}{K \cdot kg}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$


Coolant dynamic viscosity
1100
1000
900
800
700
600
500
400
300
μ [$\mu Pa \cdot s$]
−180
−150
−120
−90
−60
−30
0
30
Engine axis [mm]


Wall temperatures T_{wg} and T_{wl}
760
720
680
640
600
560
520
480
440
T [K]
Hot wall
Cold wall
Maximum allowable temperature
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]


Wall stress
σ [MPa]
350
300
250
200
150
100
50
0
−50
−100
−175
−150
−125
−100
−75
−50
−25
0
25
Engine axis [mm]
Hoop stress
Thermal stress
Maximum wall stress
Yield strength


Wall conductivity (CuCrZr)
352
348
344
340
336
332
328
324
320
316
λ_w $[\frac{W}{m \cdot K}]$
−180
−150
−120
−90
−60
−30
0
30
Engine axis $[mm]$